

## **Slovenia**

### **Competition Protection Agency**

#### **I. Introduction**

The Slovenian Competition Protection Agency ('NCA') conducted its work until recently almost completely on paperwork. However, due to recent events and the sophisticated tools that the companies and other subjects are using to automatize their services, the NCA has undertaken some steps to narrow this gap and to introduce some tools and measures to overcome the paperwork challenges and move away from paper documentation only.

In this article, two tools will be briefly described in terms of their usage, applicability, terms and conditions of usage and the practical value that the NCA gains with the applicability of these tools. With the use of these tools, the NCA has been able to automate its processes and increase its efficiency when dealing with a large number of information, external data, and the modeling of such data.

The NCA has adapted its usual way of gathering information with current ways and contemporary measures to gather information necessary for the execution of its jurisdiction, which has led to efficiency gains and a more effective approach to analyzing of data.

#### **II. 1KA AVK e-data gathering tool**

Initially, we will focus on the e-data gathering tool that the NCA has been using for online surveys, requests for information or questionnaires that the Agency submits to different stakeholders, and subsequently the handling and processing of gathered data. The NCA opted for a source application that enables services for online surveys or questionnaires. The development takes place at the Centre for Social Informatics, at the Faculty of Social Sciences, University of Ljubljana. The University of Ljubljana is also the formal owner or addressee of the corresponding intellectual property.

It is an online service that combines support for the following functionalities:

- Development, design, and technical creation of an online questionnaire;

- The implementation of online surveys: support for invitations, publication, and distribution of data;
- Compiling and analyzing data and metadata.

The basic application can be used unlimitedly and free of charge for the purposes of online surveys for basic users, under certain terms of use.

The NCA has agreed on a special module or user upgrade of the user interface on a separate domain which allows SCPA employees to create questionnaires based on a default customized questionnaire exclusively for SCPA's needs.<sup>41</sup> At the same time, the module is upgraded so that the questionnaire completion certificate and the summary of answers are forwarded to both the respondent and the SCPA head office. In doing so, the SCPA satisfied the procedural requirements of the Competition Act and the General Administrative Procedure Act. SCPA pays an annual fee for the use of 1KA AVK customized module.

#### **A. Purpose of use for the NCA**

The NCA uses the 1KA AVK service for the needs of collecting information and data from a large number of addressees, as all collected answers for each individual questionnaire are kept in a single database. SCPA employees export the data from this database and then process and analyze the data locally. 1KA AVK is therefore not used for analysis but only for faster data collection, although the analysis of data collected in such a way has proved to be much easier and accessible through different software.

#### **B. General terms**

The general terms of 1KA online survey services represent a set of conditions, which users must consider when using 1KA services. As part of the General Terms, there are also Privacy policy, Cookie policy and Anti-spam policy.

#### **C. Data Security**

The communication between users and 1KA website runs via SSL standard (Secure Socket Layer) which ensures that the data cannot be accessed by third parties

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<sup>41</sup> 1KA Enlik Anketa, <https://vprasalniki.avk.si/> (last accessed June 7, 2023).

with the help of encryption. Moreover, sensitive data, which includes user passwords in 1KA database, is stored in encrypted form.

Only addressed companies can access the SCPA specific questionnaire. Each questionnaire has a different password. Recipients receive a password with a request for information (RFI) as part of a formal request for information.

#### **D. Storage and accessibility of survey data**

1KA services on servers in Slovenia permanently store and archive all surveys for users that have exclusive rights to use data collected through their surveys. A professional provider who stores the data adheres to the highest standards in terms of power, security, etc. All the data (every click of each survey separately) is stored in parallel on the secondary server (at another location). In addition, backups are performed daily on the outdoor unit, so it is theoretically almost impossible that the data would go missing.

Server log files are archived to ensure the reliability of 1KA server and they record errors in the implementation processes. Access log files are also stored and they contain time, IP address, browser type, destination file on the server and outcome of requirement (successful, failed). Log files do not allow the identification of survey responses, as they are stored separately from the user and survey data.

### **III. Data analyzing tool**

For the purpose of more efficient analysis of different types of documents obtained during the inspections of the NCA, the NCA started to use forensic tools a few years ago when it realized that it is impossible to analyze a large number of information and data with manual tools and manual inspection of seized files. The NCA, based on the experience gathered in a number of conducted inspections, started working towards a more automated approach to the analysis of data and documents. To make the process of data gathering, analysis, and review of documents electronically accessible and eligible for an automated review using forensic IT tools, the NCA opted for a thoroughly tested IT tool called X-Ways which proved to be tailored to our needs. This tool is mainly used during inspections (on the premise inspections) and subsequent analysis of obtained documents, but occasionally also on documents that the NCA receives from the companies on request, either by RFI or by other means of information gathering (*i.e.*, during sector inquiries etc.).

### **A. Data security**

The software provides various data protection measures to ensure the acquired data is secure and confidential. Firstly, the program creates a forensic image of the original data source in order to preserve the integrity of the data. The program also provides an option to encrypt the acquired data using AES-256 encryption, which is a widely recognized and secure encryption standard. Additionally, this software provides various authentication and access control measures to ensure that only authorized employees of the NCA can access the acquired data. These measures include password protection, access permissions and user management features. Overall, the software provides comprehensive data protection measures to ensure that acquired data remains secure and confidential throughout the investigation process. During the inspections the software is used in an isolated environment (fresh install, air-gap etc.), and in the presence of the company representatives, since the company has a right to be present during acquisition and analysis of the data. During the acquisition of the data, the NCA takes special care to the fact that only relevant data is extracted, the system is wiped clean and the acquired copies are sealed until the end of proceedings, all with the aim of ensuring the safe use of the obtained data and their adequate protection. In this phase the NCA seals the data acquired in the process of inspection and it remains sealed until it is inspected and reviewed by the NCA officials in the presence of the inspected company representatives so the transparency is safeguarded.

In the process of reviewing the documents and data acquired during the inspections, the NCA separates the relevant documentation and data eligible for the proceeding to a separate folder and hands one copy of all the relevant data to the company involved in the proceeding hence their right to transparency and exercise of their right to defense of all gathered information and data is safeguarded through this procedure. After the review process, all the data not relevant to the proceeding and not used in the process of proving an infringement is deleted from all the data storage mediums.

### **B. Terms and conditions**

The NCA has no formal relation with providers of any software used and we acquire it on their publicly announced commercial terms. Used software has everything necessary for acquisition and analysis, such as data extraction (forensic copies or logical copies), verification, analysis (by use of search terms, or index),

deduplication, filtering, OCR, etc. Furthermore, some specialized tools are used in some instances (especially for email).

The terms of usage for X-Ways software are outlined in the EULA, which the NCA was obliged to accept prior to the usage. The EULA specifies the conditions under which the NCA can use the software, especially outlining the number of installations and users allowed, the types of activities that the software can be used for, and the restrictions on modifying or reverse-engineering the software.

The software was licensed to the NCA for use on a per-user basis, and the NCA was required to purchase a license for each installation of the software, which grants the right to use the software for the purpose of digital forensics. The NCA is required to comply with all applicable laws and regulations throughout the usage of the software, including data privacy laws and regulations. Failure to comply with the terms of the EULA can result in the revocation of the license.

The NCA does not perform any cloud analysis or other kinds of analysis where issues regarding unauthorized sharing of data to be analyzed or proprietary solutions contained within the software could arise.

### **C. Practical value of the software**

The data analysis of this software is performed through a variety of built-in features and tools. The program uses advanced algorithms and techniques to process the acquired data and provide insights into the contents of different sources that are discovered during on-site inspections.

One of the key features of X-Ways is its ability to create a logical view of the acquired data, which allows investigators to browse and analyze the data in a hierarchical manner. The program provides advanced search capabilities, allowing the authorized employees of the NCA to search for specific keywords, file types or metadata within the acquired data.

X-Ways also allows the NCA to identify and recover deleted files and hidden data that may not be visible through traditional browsing or searching. This can be especially useful in cases where data has been intentionally or accidentally deleted or obscured which can occur even before or during investigations of the NCA.

Overall, the data analysis is performed through a combination of advanced algorithms, search capabilities, data carving and analysis tools, allowing the NCA to

gain deep insights into the contents of the acquired data. This enables the NCA to acquire evidence of different infringements during an investigation even if those stem from different digital sources. The usage of this software has widened the investigative toolbox that the NCA has at its disposal and helped the NCA to acquire, safeguard, analyze and use data regardless of a digital source and conduct a successful investigation.

Given that companies today communicate and do business in a digital environment, the use of software for analysis of vast quantities of data is a must. We use established commercial tools that are established software in the forensic environment and suits our budget and requirements.

#### **IV. Conclusion**

As set out in this article, there is an ever-increasing need to apply tools for automation that enable the NCA to perform its tasks--especially in the field of data extraction and analysis-- to be more efficient and able to tackle the challenges in the contemporary digital environments.

It has to be mentioned that the NCA does not have a designated unit or personnel dedicated only to the use of these tools. This is the reason why the NCA is facing challenges when applying these tools as it is a voluntary act of a small group of regular case handlers of the NCA that use their spare time to perform and give trainings among the group and to tackle different challenges such as the purchase of adequate equipment and hardware, adapting to different situations during inspections and performing 'mock-up' acts to gain the necessary skills and knowledge in order to be proficient in the real-time situations.

The companies under scrutiny are more and more advanced in the digital world and use a large number of computational software, therefore, the NCA is forced to follow this trend. However, due to budgetary constraints there is a 'follow up' situation happening in this area, where the NCA aspires to be the force that has the advantage in comparison to private undertakings to tackle competition issues even more effectively with the use of advanced computational tools.